The Structural Role of Economic Complexity within the Global Trade Network

Research Questions

In this project, we examine how economic complexity influences the formation and evolution of the Global Trade Network (GTN). Building on Sonis and Hewings' (1998) argument that complexity, as an emergent property of interregional trade, has significant implications for the structural stability and transformation of economic networks, we extend the scope of analysis beyond structural dependencies to incorporate dyadic and node-level determinants of network configuration.

highligth the problem NOT the contribution

The foundation of this project lies in the work of Hidalgo et al. (2007) and Hidalgo and Hausmann (2009). In the 2007 paper they established that economic development is a path-dependent process, they have constructed the product space which has a densely connected "core" of complex, high-value goods and a sparsely connected "periphery" of simpler, low-value goods. The "innovation barrier" problem is a network problem: for a country to develop, it must move from the periphery to the core. However, countries can most easily "jump" to products that are "nearby" in this space. Because the periphery is sparsely connected to the core, most poor countries are effectively "trapped," as jumping to the high-value core would require traversing a large distance in the network. Later, Hidalgo and Hausmann (2009) have worked out the formal measure for the capabilities that allow a country to navigate this space, the Economic Complexity Index (ECI). It is a recursive measure of a country's productive knowledge, or "capabilities," derived from the diversity of products it exports and the non-ubiquity of those products. This means that ECI is a strong predictor of future growth.

better connection within the parag.

this could be the res. prob.

√ complexity – trade connections ⇒ why it matters

? include here why

At the node level, we hypothesize that countries exhibiting higher economic complexity possess a greater overall likelihood of forming trade ties. Given that such countries typically combine a more diverse array of capabilities in the production of goods (Hidalgo & Hausmann, 2009), we theorize that they may source inputs from a broader set of exporters, while simultaneously attracting a larger number of importers due to the relative non-ubiquity of their products.

too big jump here could come par 1.

At the dyadic level, we expect countries with more similar complexity scores to form a greater number of reciprocal trade ties. Higher complexity is associated with more stringent technological requirements within value chains, which implies that inputs tailored for high-complexity production processes may hold limited value for lower-technology economies, and vice versa.

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I could ~~this~~ not find this in the paper

Fagiolo et al. (2009/2010) provided some of the most robust empirical proof that the International Trade Network (ITN) is a complex network with a "small-world property, a high clustering coefficient", they have also showed that the network has a clear "core-periphery structure". They identified a "'rich club phenomenon'", their work showed that trade is concentrated among high-GDP countries, and specifically that "rich countries display more intense trade links and are more clustered". They used GDP as the attribute, in our project we would like to test if ECI is a better predictor.

APL?

why is small-world important???

At the structural level, we examine whether high-complexity countries differ systematically from low-complexity countries in the extent to which they participate in small-world network structures. If the preceding hypotheses hold, both high- and low-complexity economies should exhibit local clustering, amplified by triadic closure and network externalities; however, these mechanisms are expected to produce structurally different outcomes for the two groups.

same thing

For high-complexity economies, triadic closure is likely to occur among globally central and technologically advanced partners. Because these partners tend to maintain diverse and far-reaching trade portfolios, closure among them contributes to a dense and highly interconnected core. Network externalities in this segment of the network thereby reinforce global reach, increasing the probability that any two high-complexity countries can be connected through a small number of intermediaries.

Low-complexity economies, by contrast, may also experience triadic closure, but predominantly within more regionally confined or technologically homogeneous clusters. These clusters, while locally dense, tend to be less integrated into the wider trade system and generate weaker global externalities. As a result, their clustering does not translate into the same degree of long-range connectivity.

Consequently, high-complexity economies should exhibit shorter average path lengths due to their position within a globally interconnected core, whereas low-complexity economies - despite local clustering - remain more peripheral because their ties link them primarily to partners who are themselves weakly connected to the broader network.

Data Collection

Can you think about a "causal" story of play involving? trade factors homophily distance

Descriptive is nice but difficult to apply.

The project requires three distinct types of data to construct and analyze the GTN for a single cross-sectional year. These three types are:

We will use Bilateral Trade data from the UN Comtrade Database to build the GTN, where the nodes will represent the countries and the edges the trade links. citation

After that, our plan is to calculate ECI by ourselves based on Hidalgo et al. (2007) and Hidalgo and Hausmann (2009). To that we will use Harvard Growth Lab: Atlas of Economic Complexity database, using Country-Product Exports to build the Mcp matrix, Total World Product Trade and Total Country Exports to calculate RCA. OEC data?

We will also need control variables, like GDP, population, trade (% of GDP) and political stability for our ERGM model which we plan to download from the World Bank Open Data. introduce first

Methodology

- what time frame of trade?
- aggregate or dynamic?

As a measure of economic complexity, we adopt the method of reflections proposed by Hidalgo and Hausmann (2009). Equations [1] and [2] summarize the two types of variables derivable from this approach:

$$k_{c,N} = \frac{1}{k_{c,0}} \sum_{p} M_{cp} k_{p,N-1} \qquad [1]$$

$$k_{p,N} = \frac{1}{k_{p,0}} \sum_{c} M_{cp} k_{c,N-1} \qquad [2]$$

where M_{cp} is an adjacency matrix, where if country c is a significant exporter of product p and 0 otherwise. Whether a country is a significant exporter of product p is determined by whether its Relative Comparative Advantage (RCA) for the product exceeds 1, following Hidalgo et al. (2007). RCA measures whether a country exports more of product p, as a share of its total exports, than the average country.

The variable $k_{0,c}$ represents the diversification of a country in terms of the number of products the country is a significant exporter of, while $k_{p,0}$ represents the ubiquity of a product in terms of the number of countries that are significant exporters of it. We plan to utilize higher-order iterations of the $k_{c,N}$ variable with even values of N. These higher-order measures can be interpreted as the average diversification of a country's neighbors, weighted by the probability that a random walker originating from a given node reaches another node after N steps. Conceptually, this reflects the expected diversification of other countries' capabilities as embedded within the export structure of country c. Increasing the order N allows the measure to increasingly abstract from simple quantitative features, such as trade volume, thereby isolating the structural component of economic complexity.

We intend to test the node- and dyadic-level hypotheses using Exponential Random Graph Models (ERGMs). This approach estimates the effects of structural, dyadic, and node-level attributes on the likelihood of observing a given network. ERGMs employ Markov Chain Monte Carlo Maximum Likelihood Estimation (MCMC-MLE), which simulates numerous random networks and iteratively adjusts attribute coefficients to maximize the probability of reproducing the observed network (J van Der Pol, 2019). Positive coefficient estimates indicate that nodes possessing the corresponding attributes are more likely to form ties, conditional on the rest of the model.

why is ERGM good and what are the limitations?

Setayesh et al. (2022) and analyzed the global trade network using Exponential Random Graph Models (ERGMs) and Separable Temporal ERGMs (STERGMs) to understand the structural, economic, geographical, political, and cultural factors influencing trade relations. They have found that reciprocity and triangle formations significantly impact trade relations. Countries tend to trade in both directions and form triangular trade patterns. The study found structural, economic, geographical, and political factors to be significant in shaping the GTN. They have used the gravity model as a baseline for comparison with the ERGM findings, and the study found alignment between the two. De Benedictis & Tajoli (2011) explicitly state their goal is to use network analysis to "complement some of the empirical findings of the gravity models" and that their "network indices are also used in a gravity model regression", which means they did not treat network analysis and gravity models as separate.

We also explored the potential use of feature learning techniques and link prediction models based on similarity measures, such as the Jaccard similarity, following approaches applied to social media networks (e.g., Thai et al., 2016). However, we ultimately opted for ERGMs, as evaluating the performance of machine learning methods that rely on labeled data would necessitate withholding a portion of our observations as a test set. Given the limited number of countries for which trade data are available, we deemed such a split to entail a substantial loss of information.

For a measure of how similar a network is to a small world we are going to deploy the Small-World-Ness Score used by Humphries & Gurney (2008). The score rewards high global clustering coefficients and short average path lengths, which are the primary characteristics of small-world networks. We will construct two separate country-level networks: one representing high-complexity countries and one representing low-complexity countries' GTN. Countries will be ranked according to ECI, after which we will select an equal number of countries from the upper and lower ends of the distribution. Countries in the middle of the distribution will be intentionally excluded to maximize contrast between the two groups. For both networks G_1 and G_2, we will compute a clustering coefficient (C_G) using the transitivity-based definition (ratio of triangles to connected triples), an average shortest path length (L_G) and the equivalent metrics (C_{random} and L_{random}) obtained from an ensemble of Erdős-Rényi random graphs with the same number of nodes and edges. Small-worldness will then be calculated as indicated in Equation [3]:

$$S = \frac{C_G/C_{rand}}{L_G/L_{rand}} \qquad [3].$$

To formally assess whether each of these networks differ statistically significantly in their small-world structure (whether the high-complexity network is more small-world than the low-complexity network), we will apply the bootstrap-based small-worldness test proposed by Lovekar, Sengupta, and Paul (2024). We will implement a non-parametric bootstrap procedure. This is necessary as the S-statistic is a ratio of point estimates, and its analytical distribution is unknown. First, we will generate a large ensemble of random networks to serve as a null model for each of the two empirical networks. Next, we will generate two empirical distributions of S-scores by conducting bootstrap iterations. In each iteration, we will resample (with replacement) the statistics C_{random} and L_{random} from the null models to compute a single bootstrap replicate of S. This process yields two distributions of S-values, which represent the sampling distribution of the S-score for each network. Finally, these two distributions will be formally compared using a Mann-Whitney U test to test the null hypothesis that the two S-scores are drawn from the same underlying distribution.

Expected outcomes

As outlined in our hypotheses, we anticipate that countries with higher economic complexity scores will exhibit a greater likelihood of forming ties. This expectation would be reflected in a statistically significant positive coefficient for economic complexity in the ERGM estimates. Similarly, the hypothesis that countries with similar levels of economic complexity are more likely to be connected would be supported by a statistically significant negative

? network measures will be biased

there is low-high EC trade

coefficient for the difference in complexity scores between two countries. Non-significant findings, however, may arise due to the substantial explanatory power of standard structural controls, such as the presence of triangles or centrality measures, as well as country-level covariates including GDP growth, population density, and trade volume.

The hypothesis that networks of high-complexity countries exhibit a greater resemblance to small-world structures than those of low-complexity countries would be supported by non-overlapping confidence intervals around their respective average small-world-ness scores, obtained via bootstrapping. An alternative explanation for contrary results could be that high- and low-complexity countries each form highly clustered subgroups, which are interconnected through weaker ties, producing a small-world structure at the aggregate network level.

what is the relevance of this?

Impact

The goal of this research is to have a significant impact by providing a new, knowledge-based paradigm for understanding the global economy. Bridging ECI and ERGMs has not been extensively examined before so the project would result in one of the first generative models of the Global Trade Network that can simultaneously test the effects of productive capabilities (ECI), economic size (GDP), and endogenous network structures like triadic closure.

why is triadic closure in aggregate trade?

The primary impact would be to empirically demonstrate how development traps are formed and maintained, showing that the global core-periphery structure is not just a "rich club" but a "capability club".

This research would be useful for national and international policymakers, such as governments because it helps them identify a country's specific "binding constraints" to growth and target investments to help peripheral regions escape development traps. International economic institutions like the World Bank, IMF, and OECD, which are already embracing economic complexity principles, and could also benefit from using this model.

The results could be used by corporate and supply chain leaders, for example by helping smaller firms identify new market opportunities or find new suppliers by looking for partners with compatible technological alignment, not just low prices. The model would be useful for mapping and understanding vulnerabilities in global value chains (GVCs), because it would help companies move from a "minimize cost" logic to a "cost of resilience" mindset by identifying single-supplier dependencies.

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